


What is Machine Learning?

ARTIFICIAL INTELLIGENCE VS MACHINE LEARNING VS DEEP LEARNING

1 Artificial Intelligence

Development of smart systems and machines that can carry out tasks that typically require human intelligence

2 Machine Learning

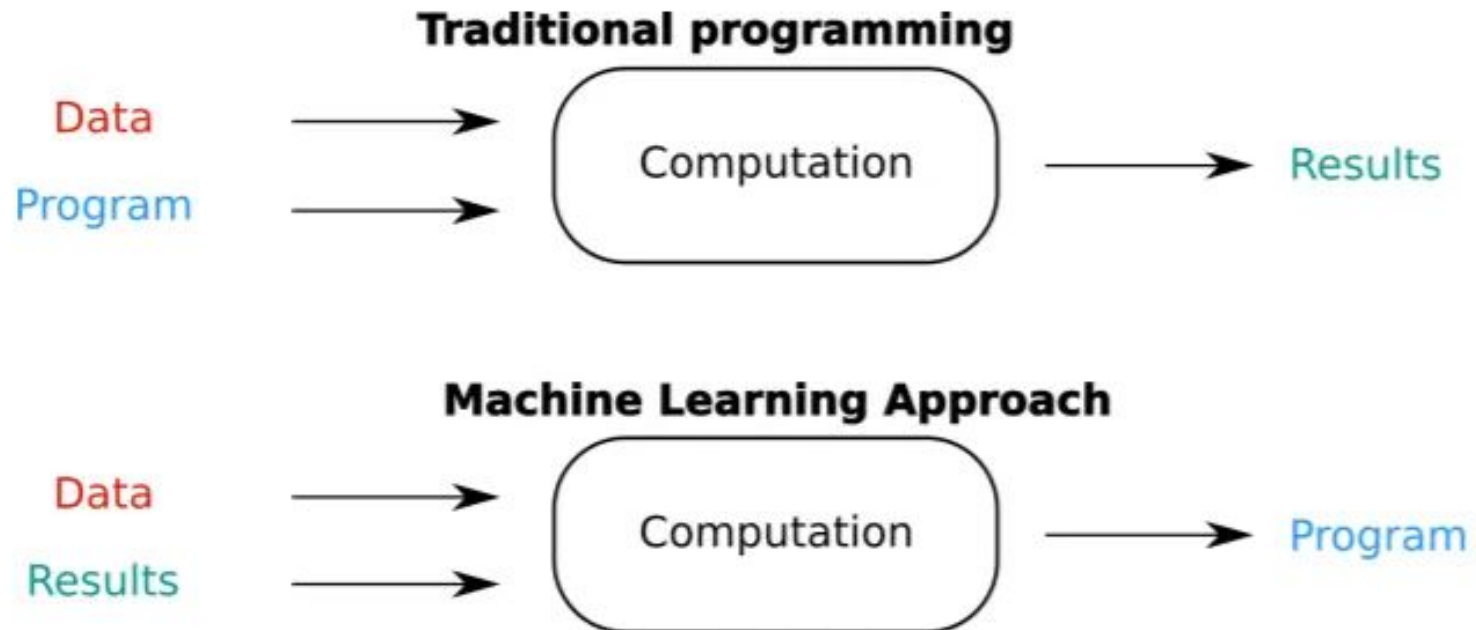
Creates algorithms that can learn from data and make decisions based on patterns observed
Require human intervention when decision is incorrect

3 Deep Learning

Uses an artificial neural network to reach accurate conclusions without human intervention

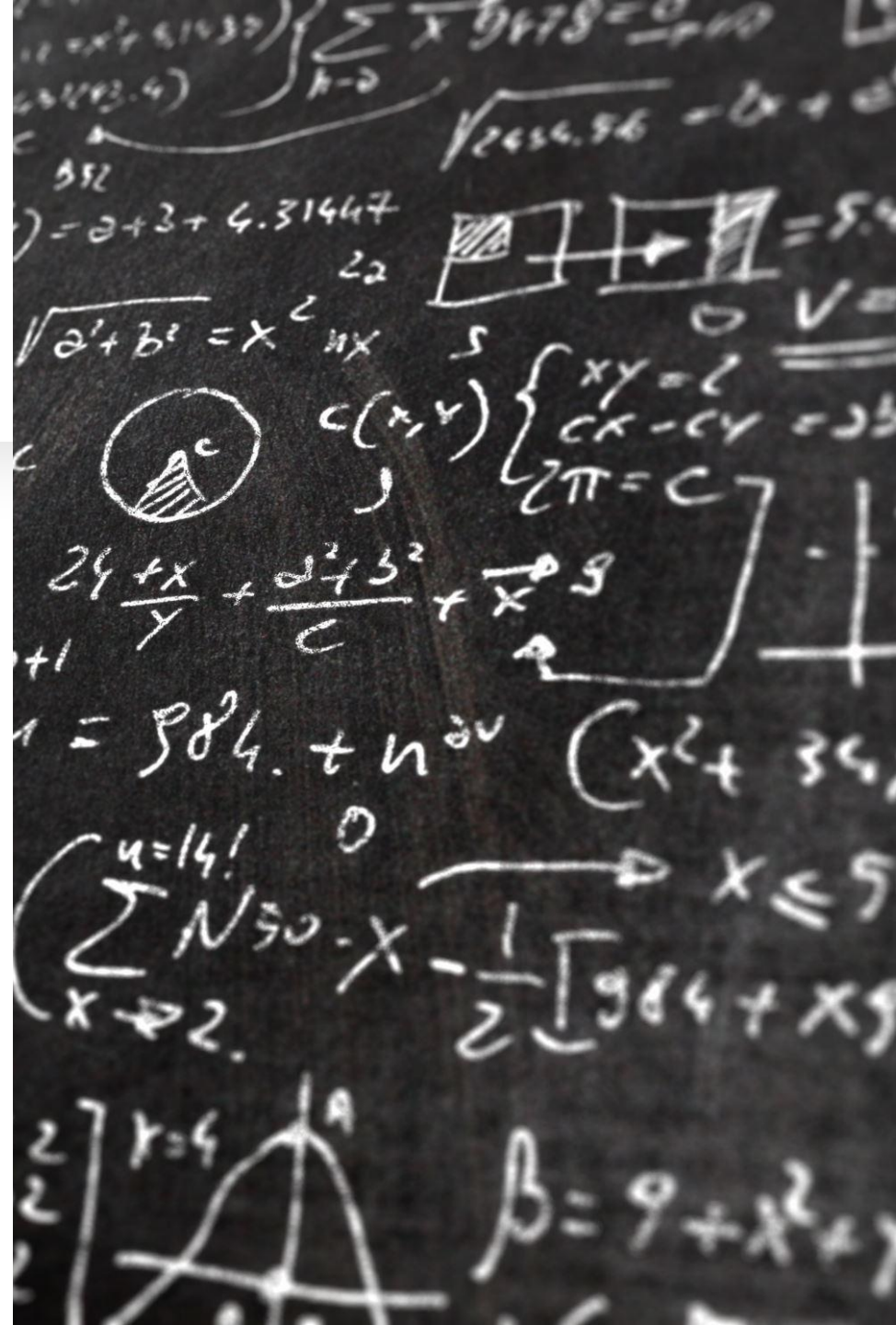
What is Machine Learning?

- Machine Learning vs. traditional programming

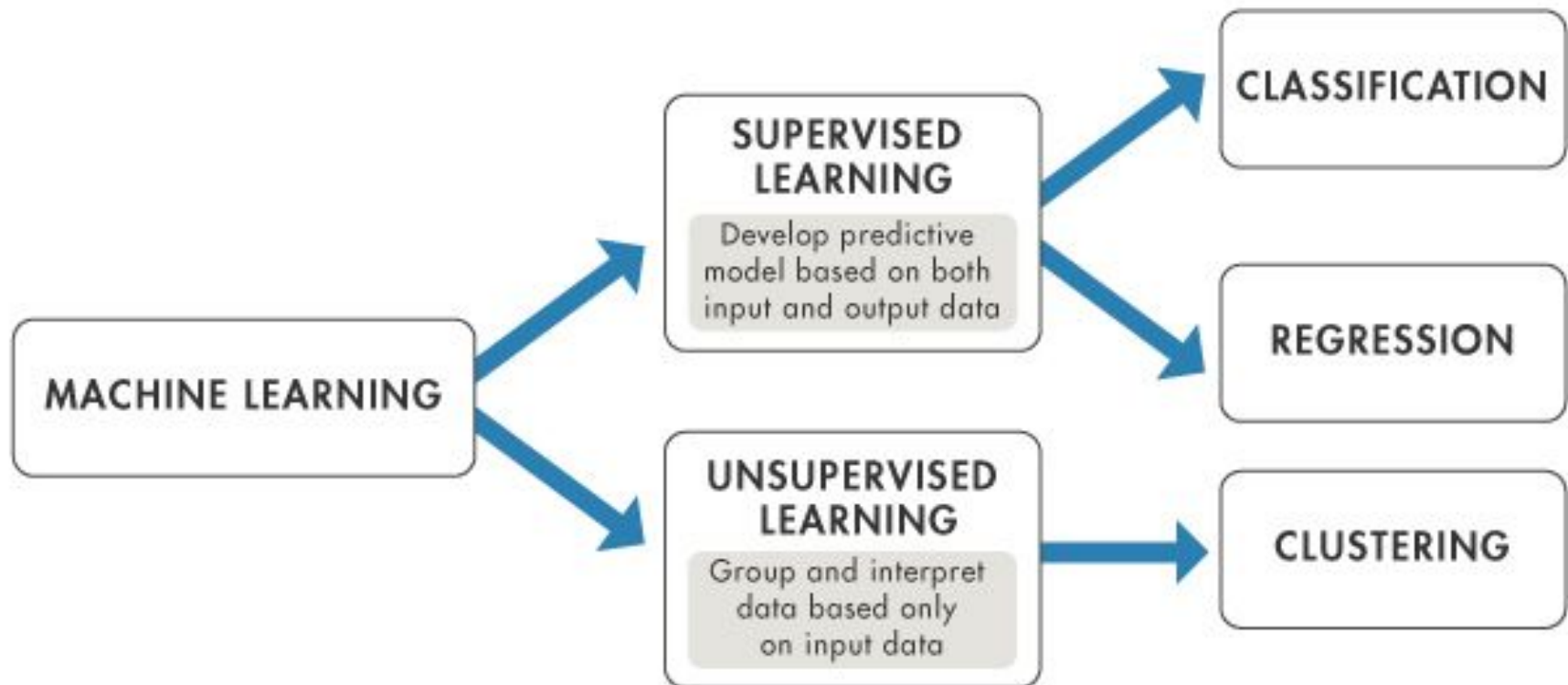


What is Machine Learning?

- Learning patterns from data
- Used to make predictions or find structure
- Goal: perform well on new, unseen data



Types of Machine Learning

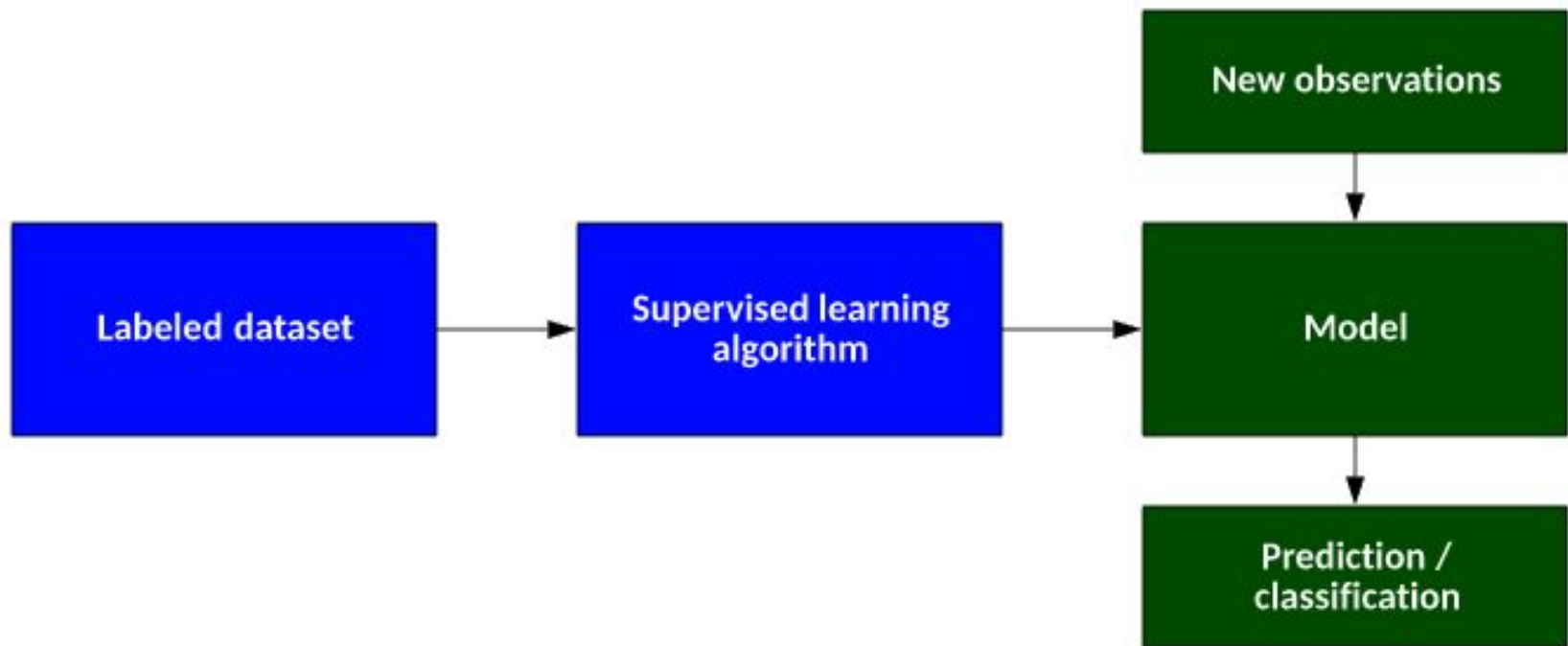


Supervised Learning

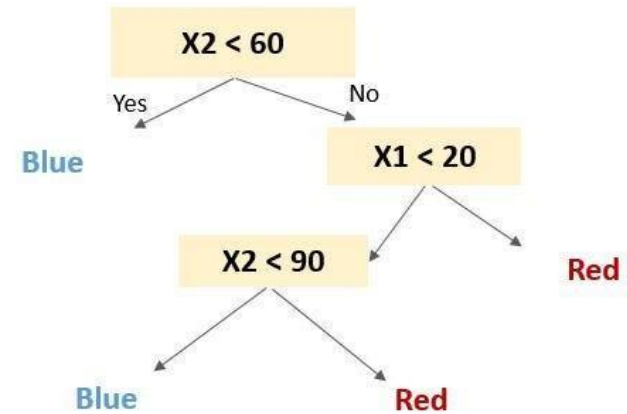
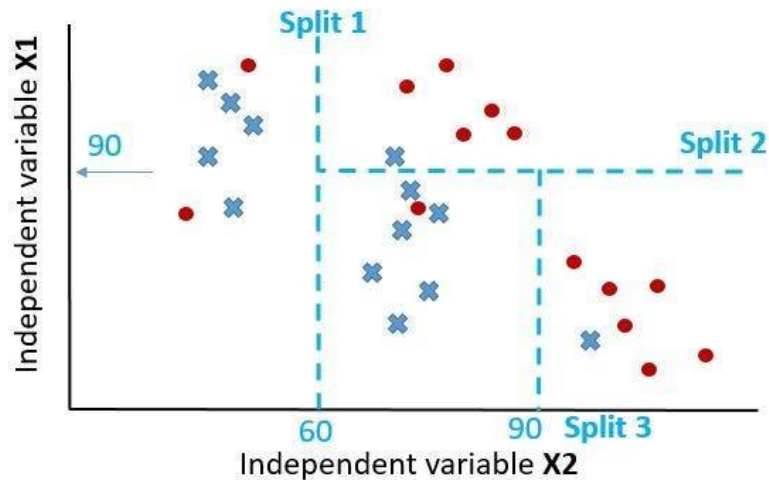
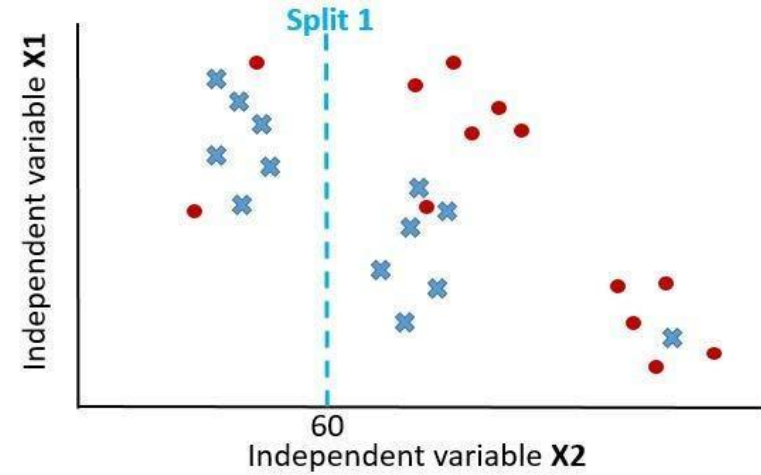
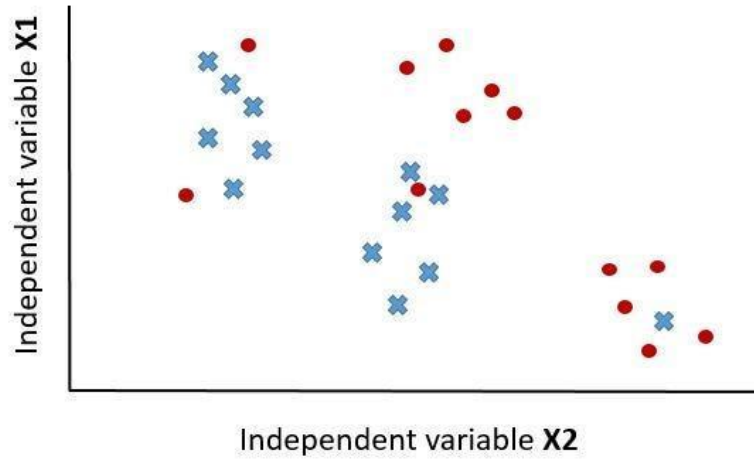
Supervised learning learns from examples with known answers.

Regression or classification.

- *house prices*
- *spam detection*
- *churn prediction*



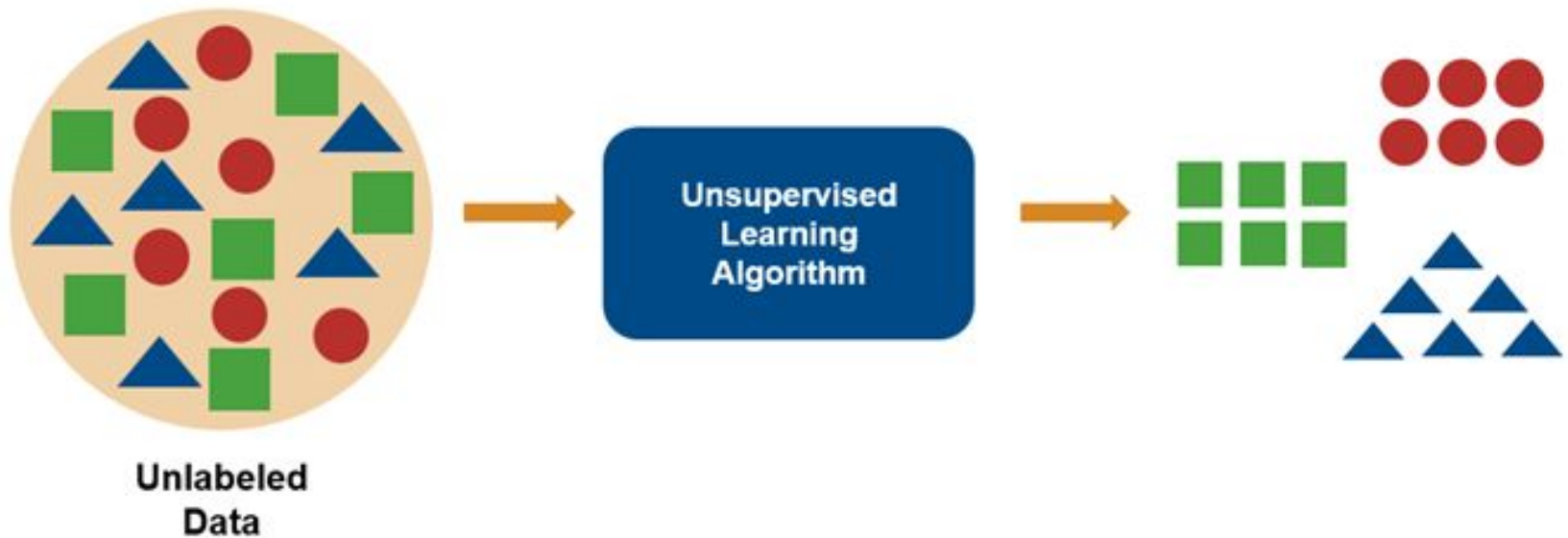
Decision Tree



Unsupervised Learning

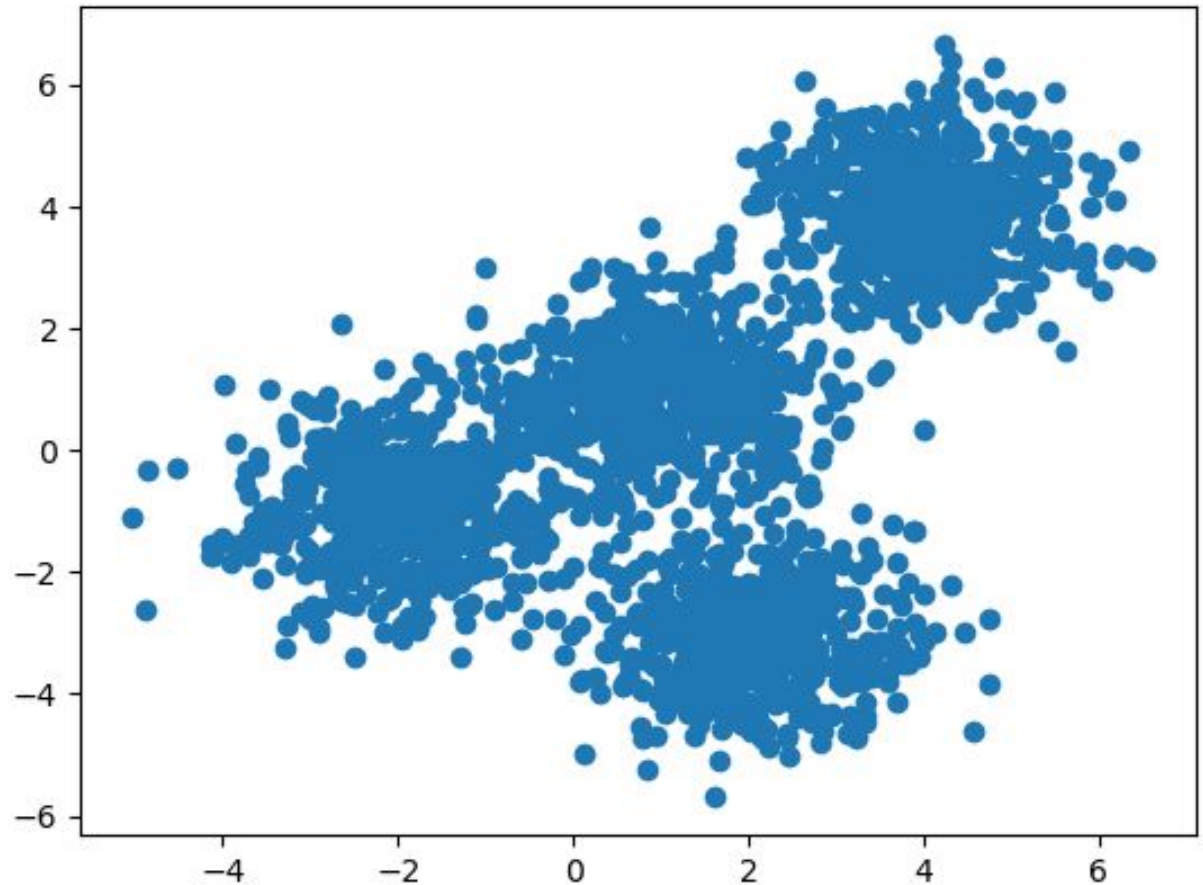
Unsupervised learning looks for patterns without being told the answer.

- Clustering items
- Customer segmentation



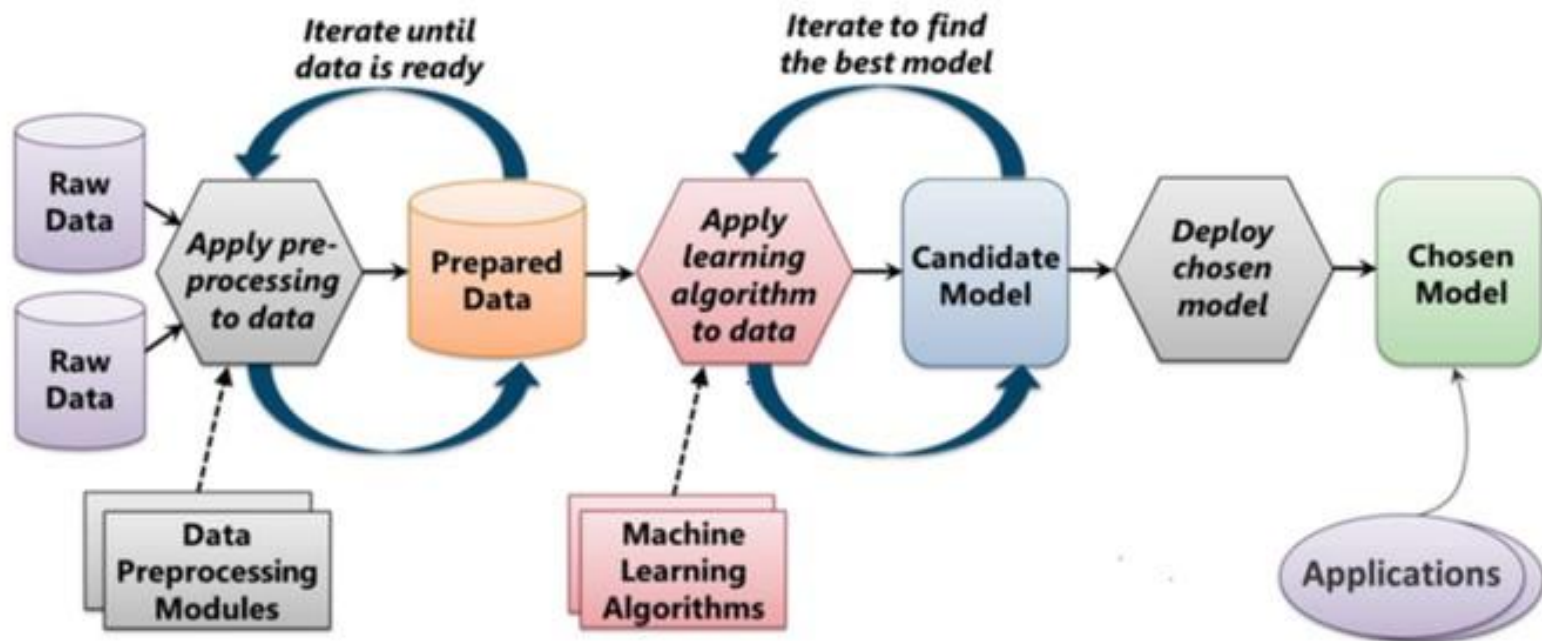
K Means Model

- Random centroids
- Find points close to each centroid
- Find mean of each cluster as new centroids
- Repeat until no change



Typical Machine Learning Workflow

- Collect / prepare data
- Choose and train a model
- Evaluate model performance
- Improve and iterate
- Deploy



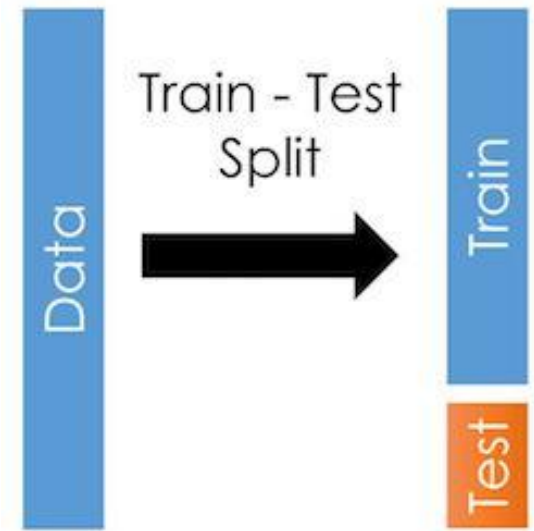
Train/Test Split

- Split data into 2 (or 3) sets.
- Test dataset is reserved for final evaluation to make inferences on unseen examples.
- Avoids overfitting and misleading performance metrics

scikit-learn:

```
from sklearn.model_selection import train_test_split

# Split the data with 33% for testing
X_train, X_test, y_train, y_test = train_test_split(
    X, y,
    test_size=0.33,
    random_state=42
)
```



Baseline

- What accuracy do we get selecting classes randomly?
- What if we assume all are of the majority class?
- Generate a baseline to compare with our actual models.
- We want a model that performs better than by chance!

Scikit-learn:

```
from sklearn.dummy import DummyClassifier  
  
clf = DummyClassifier(strategy="most_frequent")  
clf.fit(X_train, y_train)  
clf.score(X_test, y_test)
```



Model Selection

- “Spaghetti method”: throw many models at the data and see what sticks. (Not recommended.)
- Select possible models from EDA. For example, if classes are not readily linearly separable, consider tree models or even neural networks.



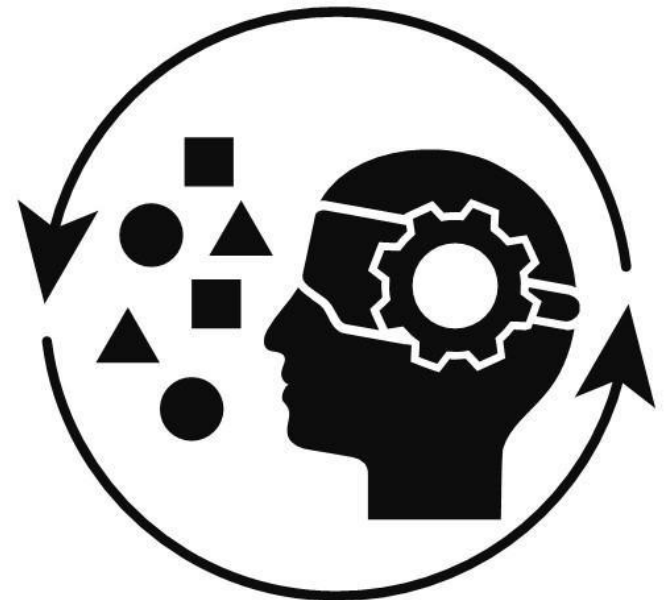
Train a model

- Training models (scikit-learn always uses same pattern).
- Get performance metrics (confusion matrix and classification report).

scikit-learn:

```
from sklearn.tree import DecisionTreeClassifier
from sklearn.metrics import confusion_matrix, classification_report

clf = DecisionTreeClassifier()
clf.fit(x_train, y_train)
y_pred = clf.predict(X_test)
print(confusion_matrix(y_test, y_pred))
print(classification_report(y_test, y_pred))
```



How Do We Know If a Model Is Good?

Training accuracy is not enough

Need to evaluate on new, unseen data

Common ideas: accuracy, confusion matrix, precision, recall

Key goal: generalizability

Confusion Matrix

- Not a band (though it'd be a great name for one)
- A performance evaluation table for machine learning classification models, comparing predicted classes against actual target values

		Predicted Values	
		Positive	Negative
Actual Values	Positive	TP	FN
	Negative	FP	TN

Classification Report

- Classes
- Metrics for each class – precision, recall, f-1 score
- How many of each class?
- Accuracy
- Averages of individual metrics

The diagram illustrates the components of a Classification Report. It features a table with five main sections, each highlighted with a colored border and a numbered callout:

- 1 (Yellow):** Class Name (y_true, y_pred)
- 2 (Purple):** Each Class Metrics (precision, recall, f1-score)
- 3 (Blue):** Actual instance of each class (support)
- 4 (Orange):** Overall Accuracy (accuracy)
- 5 (Pink):** Metrics Average (macro avg, weighted avg)

	precision	recall	f1-score	support
y_true	0.67	0.80	0.73	5
y_pred	0.75	0.60	0.67	5
accuracy			0.70	10
macro avg	0.71	0.70	0.70	10
weighted avg	0.71	0.70	0.70	10

Class Name

Each Class Metrics

Actual instance of each class

Overall Accuracy

Metrics Average

Metrics

- Accuracy
- Precision
- Recall
- F-1 Score

$$Accuracy = \frac{TP}{TP+FP+TN+FN}$$

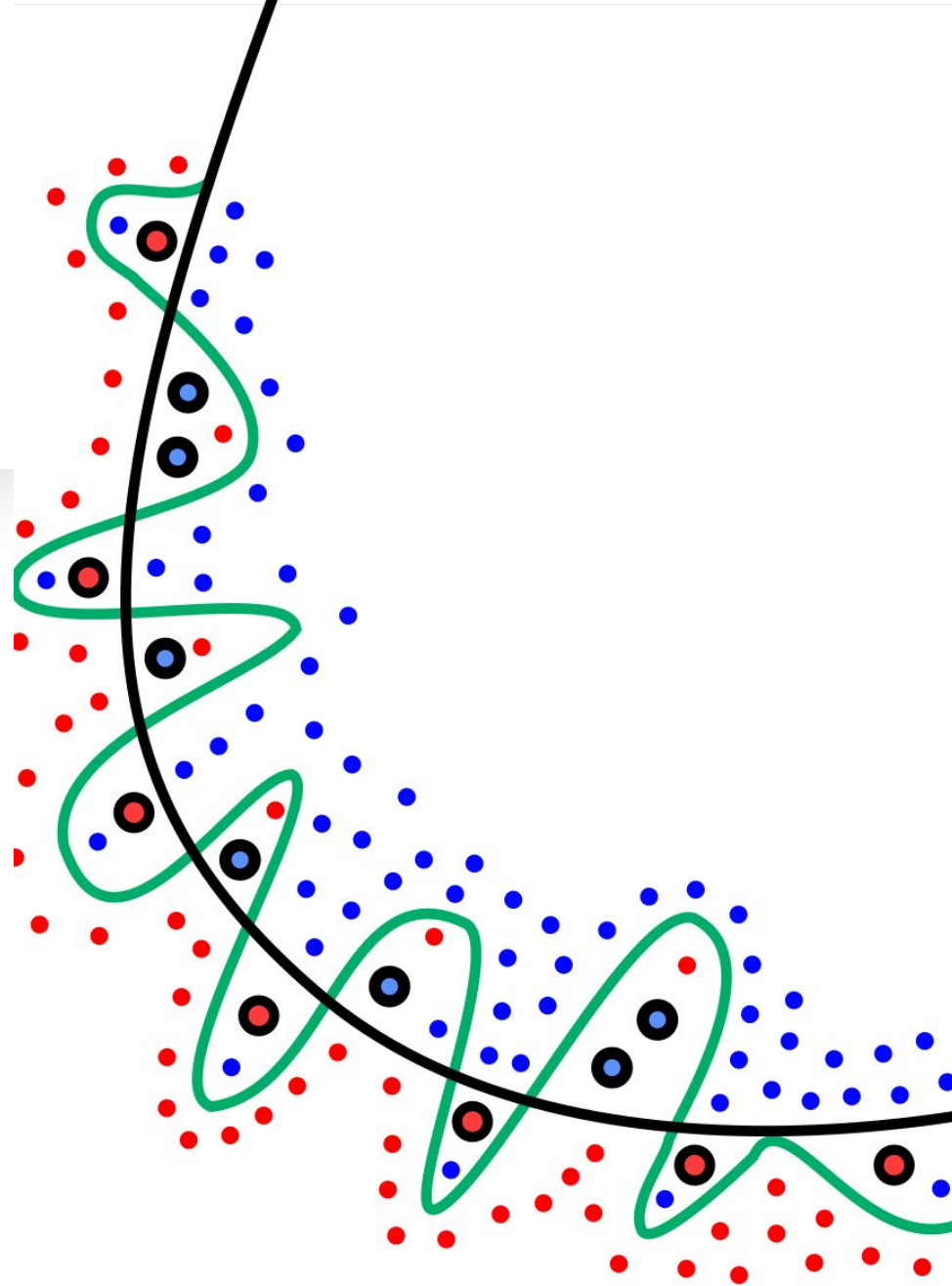
$$Precision = \frac{TP}{TP+FP}$$

$$Recall = \frac{TP}{TP+FN}$$

$$F1 = 2 \frac{Precision * Recall}{Precision + Recall}$$

Generalization and Overfitting

- Overfitting: model learns training data too closely
- Cross-validation helps test reliability
- Regularization helps keep models from becoming too complex
- We want models that capture real patterns, not just noise



A photograph of a wooden beam balanced on a single dark sphere, symbolizing imbalance. The beam is horizontal and the sphere is positioned directly beneath its center. The background is a plain, light-colored wall.

Addressing Class Imbalance in Data

Definition

Class imbalance occurs when some classes have significantly fewer examples than others, leading to biased model performance.

Impact on Models

Models trained on imbalanced data may perform poorly on minority classes, resulting in lower accuracy and unfair outcomes.

Techniques for Balance

Methods like oversampling, undersampling, and synthetic data generation help create more balanced datasets for fairer learning.

Thank you!
And happy modelling!

